

CDC Pipeline Load Test Report

15 May 2026 | AWS EKS (us-east-2) | 10,000 transactions | 2s batch interval | Duration: ~199s

10,000
Records Generated

10,000
Records Delivered (100%)

2.44s
Median E2E Latency

0
Lost / Duplicates

Oracle XE 21c → Debezium CDC (LogMiner) → Kafka (Strimzi 3.8.1) → Synapse Consumer → PostgreSQL (AWS RDS)

Latency Results (seconds) — all 10,000 records

Metric	Capture (Oracle → Kafka)	Processing (Kafka → PG)	End-to-End (Oracle → PG)
Minimum	0.22	0.01	0.42
Average	2.04	0.32	2.36
P50 (Median)	2.23	0.32	2.44
P95	3.28	0.56	3.65
P99	3.34	0.62	3.79
Maximum	3.38	0.70	3.92

Percentiles use PostgreSQL **PERCENTILE_CONT** (continuous interpolation) over all 10,000 latency values. P50 = median (50% faster, 50% slower). P95/P99 = only 5%/1% of records exceed this value. Latency = EXTRACT(EPOCH FROM (end_timestamp – start_timestamp)) in seconds per record.

Infrastructure Resources

Component	Spec
Oracle XE 21c	StatefulSet, 1 pod, PDB: ACCESSBANK
Debezium CDC	3 KafkaConnectors (LogMiner), 1 Connect pod
Kafka (Strimzi)	3 brokers, 9 topics, 4 partitions each
Synapse Main	3 replicas, CDC consumer enabled
Synapse Consumer	1 dedicated CDC replica
PostgreSQL	AWS RDS (SSL), db: synapse
Karpenter Nodepool	64 vCPU limit, 2 nodes

Key Findings

- **100% delivery** with zero loss and zero duplicates
- **Synapse is NOT a bottleneck** — 0.32s P50 processing, near-instant after Kafka
- **~94% of latency is in Debezium LogMiner** — polling-based redo log scanning is the dominant cost
- **Tight P50→P99 spread** (2.44s → 3.79s) = stable, predictable, SLA-ready
- **2.4s is invisible to users** — human navigation from transfer to statement takes 3–5s

LogMiner Now → GoldenGate Later

	LogMiner (Current)	XStream (Future)
Mechanism	Poll redo logs	Push-based stream
P50 E2E	2.44s	< 0.7s projected
License	Free / open-source	GoldenGate required
Migration	1 config change, 0 app code changes	

All timestamps UTC | Tracked via 3 columns: ora_created_at (Oracle commit), kafka_published_at (Debezium publish), created_at (PG insert) | 10 filtered accounts, 4 tables, 3 connectors